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Effects of a 6-Week Repeated-Sprint Training With Voluntary Hypoventilation at Low and High Lung Volume on Repeated-Sprint Ability in Female Soccer Players

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Purpose: To investigate the effects of repeated-sprint training with voluntary hypoventilation at low (RSH-VHL) and high (RS-VHH) lung volume on repeated-sprint ability (RSA) in female athletes. **Methods:** Over a 6-week period, 24 female soccer players completed 12 sessions of repeated 30-m running sprints with end-expiratory breath holding (RSH-VHL, n = 8), end-inspiratory breath holding (RS-VHH, n = 8), or unrestricted breathing (RS-URB, n = 8). Before and after training, a running RSA test consisting of performing 30-m all-out sprints until exhaustion was implemented. **Results:** From before to after training, the number of sprints completed during the RSA test was increased in both RSH-VHL (19.3 [0.9] vs 22.6 [0.9]; $P < .01$) and RS-VHH (19.3 [1.5] vs 20.5 [1.7]; $P < .01$) but not in RS-URB (19.4 [1.3] vs 19.5 [1.7]; $P = .67$). The mean velocity and the percentage decrement score calculated over sprints 1 to 17 were, respectively, higher (82.2% [1.8%] vs 84.6% [2.1%] of maximal velocity) and lower (23.7% [3.1%] vs 19.4% [3.2%]) in RSH-VHL ($P < .01$), whereas they remained unchanged in RS-VHH and RS-URB. The mean arterial oxygen saturation recorded during training at the end of the sprints was lower in RSH-VHL (92.1% [0.4%]) than in RS-VHH (97.3% [0.1%]) and RS-URB (97.8% [0.1%]). **Conclusions:** This study shows that female athletes can benefit from the RSH-VHL intervention to improve RSA. The performance gains may have been limited by the short sprinting distance with end-expiratory breath holding, which provoked only moderate hypoxemia. The increase in the number of sprints in RS-VHH seems to show that factors other than hypoxia may have played a role in RSA improvement.